

Total No. of Questions : 12]

SEAT No. :

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[5561]-145

[Total No. of Pages : 3

B.E. (Mechanical Engineering)

INDUSTRIAL ENGINEERING

(2012 Course) (Semester - II) (Elective - III) (402049C)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer should be written in one answer book.*
- 2) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11, or Q.12.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Figures to the right side indicate full marks.*
- 5) *Assume suitable data, if necessary.*

Q1) a) Discuss functions of management? **[6]**

b) Explain briefly the productivity for manufacturing industry and discuss factors affecting the productivity. **[4]**

OR

Q2) a) Explain in detail. **[6]**

i) Span of control,

ii) Delegation of authority

b) Write a short note on "Line and Staff organizational structure? **[4]**

Q3) a) Define method study. Explain the procedure for method study. **[6]**

b) What are Therblings give any four Therbligs with Symbols? **[4]**

OR

Q4) a) What is performance rating? Explain various methods of Performance rating? **[6]**

b) Explain Two handed process Chart with suitable example. **[4]**

P.T.O.

Q5) a) Describe (any two concepts of) following with suitable example: [6]

- i) SIMO chart
- ii) Flow process chart
- iii) MOST
- iv) Qualified worker

b) Following data shows the time taken during a time study of a process. The time given is cumulative. It total allowance is 16%, calculate standard time of the process. [4]

Activity	Observed time (Min)	Performance rating (%)
1	0.24	80
2	1.28	90
3	2.32	100
4	3.15	90
5	4.48	110

OR

Q6) a) Explain in detail process of Work Sampling. [6]

b) Explain PMTS? [4]

Q7) a) Discuss objectives and functions of production planning and control. [8]

b) What is Master Production Schedule (MPS)? Discuss objective and importance of MPS. [7]

OR

Q8) a) Discuss different phases in ERP implementation life cycle. [7]

b) Explain: What is capacity planning with suitable examples? Explain briefly capacity planning process. [8]

Q9) a) Explain various factors which affect the Plant Layout planning. Enlist principles of good Plant Layout. [8]

b) Explain types of costs associated with Inventory. [7]

A company consumes 12,000 units of an item per year. Its production capacity is (60 units) per day. The cost of each unit produced by the company is Rs. 8/- The set-up and tooling cost is Rs. 96/- per setup. The carrying charges are 15% of the cost per unit. Calculate:

- i) Economic quantity to be manufactured in a branch
- ii) Frequency of production runs
- iii) Production period

OR

Q10)a) Enlist different computer aided layout design techniques used in practice. And write a short note on [8]

- i) Combination layout
- ii) Line layout

b) Explain 'ABC-Analysis' of Inventory Control. [7]

A manufacturer has to supply his customers 3600 units of his products per year. Shortage is not permitted. Inventory carrying cost amounts Rs 1.2 per unit per annum. The setup per run is Rs. 80 find.

- i) Economic order quantity
- ii) Optimum number of orders per annum
- iii) Average annual inventory cost (minimum).

Q11)a) Write a short note on depreciation and causes of depreciation. [4]

b) Explain objectives of cost accounting? Explain elements of cost. [6]

OR

Q12)a) Explain the term : KRA [6]

b) Calculate B.E.P. (In Unit and In Rupee) from the following information [4]

Fixed cost = Rs 1200/-

Variable cost = Rs 4000/-

Sales in rupee = Rs 7000

Sales in unit = Rs 1000/-

